

Calculus with Parametric and Polar Curves

ANSWER KEY



Parametric derivatives and arc length

- 1 For the curve $x = t^2$, $y = t^3$, find $\frac{dy}{dx}$ in terms of t .

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{3t^2}{2t} = \frac{3t}{2} \text{ (for } t \neq 0\text{)}.$$

- 2 Find the arc length of the curve $x = 3t^2$, $y = 2t^3$ for $0 \leq t \leq 2$.

$$\frac{dx}{dt} = 6t, \frac{dy}{dt} = 6t^2. \\ L = \int_0^2 \sqrt{36t^2 + 36t^4} dt = \int_0^2 6t\sqrt{1+t^2} dt = [2(1+t^2)^{3/2}]_0^2 = 2(5^{3/2} - 1) = 10\sqrt{5} - 2.$$

Polar area

- 3 Find the area enclosed by one loop of the rose $r = 3\sin 2\theta$ (for $0 \leq \theta \leq \pi/2$).

$$A = \frac{1}{2} \int_0^{\pi/2} 9\sin^2 2\theta d\theta = \frac{9}{2} \int_0^{\pi/2} \frac{1 - \cos 4\theta}{2} d\theta = \frac{9}{4} \left[\theta - \frac{\sin 4\theta}{4} \right]_0^{\pi/2} = \frac{9\pi}{8}.$$

- 4 Find the area enclosed by the cardioid $r = 2 + 2\cos \theta$.

$$A = \frac{1}{2} \int_0^{2\pi} (2 + 2\cos \theta)^2 d\theta = \frac{1}{2} \int_0^{2\pi} (4 + 8\cos \theta + 4\cos^2 \theta) d\theta. \text{ The } \cos \theta \text{ term integrates to 0 over a full period, and } \int_0^{2\pi} 4\cos^2 \theta d\theta = 4\pi: A = \frac{1}{2}(8\pi + 4\pi) = 6\pi.$$

Second derivatives and concavity for parametric curves

- 5 For $x = t^2$, $y = t^3 - 3t$, find $\frac{d^2y}{dx^2}$ in terms of t , using $\frac{d^2y}{dx^2} = \left[\frac{d}{dt} \left(\frac{dy}{dx} \right) \right] \div \frac{dx}{dt}$.
 $\frac{dy}{dx} = \frac{3t^2 - 3}{2t}$. Differentiating with respect to t : $\frac{d}{dt} \left(\frac{3t^2 - 3}{2t} \right) = \frac{(6t)(2t) - (3t^2 - 3)(2)}{4t^2} = \frac{12t^2 - 6t^2 + 6}{4t^2} = \frac{6t^2 + 6}{4t^2}$. Dividing by $\frac{dx}{dt} = 2t$: $\frac{d^2y}{dx^2} = \frac{6t^2 + 6}{8t^3} = \frac{3(t^2 + 1)}{4t^3}$.

Area between polar curves

- 6 Find the area inside the circle $r = 3$ and outside the circle $r = 2$ (an annular region).

$$A = \frac{1}{2} \int_0^{2\pi} (3^2 - 2^2) d\theta = \frac{1}{2}(5)(2\pi) = 5\pi.$$

Horizontal and vertical tangents on parametric curves

- 7 For $x = t^2 - 4$, $y = t^3 - 3t$, find the value(s) of t where the tangent line is horizontal ($\frac{dy}{dt} = 0$, $\frac{dx}{dt} \neq 0$).
 $\frac{dy}{dt} = 3t^2 - 3 = 0$ gives $t = \pm 1$. Check $\frac{dx}{dt} = 2t \neq 0$ at both: valid. Horizontal tangents at $t = 1$ and $t = -1$.

- 8 For the same curve, find the value(s) of t where the tangent line is vertical ($\frac{dx}{dt} = 0$, $\frac{dy}{dt} \neq 0$).
 $\frac{dx}{dt} = 2t = 0$ gives $t = 0$. Check $\frac{dy}{dt} = 3(0)^2 - 3 = -3 \neq 0$: valid. Vertical tangent at $t = 0$.

Converting between polar and Cartesian for calculus

- 9 Find the slope of the tangent line to $r = 1 + \sin \theta$ at $\theta = 0$, using $x = r \cos \theta$, $y = r \sin \theta$ as parametric equations in θ .

$x = (1 + \sin \theta) \cos \theta$, $y = (1 + \sin \theta) \sin \theta$. $\frac{dx}{d\theta} = \cos \theta \cos \theta - (1 + \sin \theta) \sin \theta = \cos^2 \theta - \sin \theta - \sin^2 \theta$;
at $\theta = 0$: $1 - 0 - 0 = 1$. $\frac{dy}{d\theta} = \cos \theta \sin \theta + (1 + \sin \theta) \cos \theta$; at $\theta = 0$: $0 + 1 = 1$. Slope $= \frac{dy/d\theta}{dx/d\theta} = \frac{1}{1} = 1$.

Speed and length for parametric motion

- 10 A particle moves according to $x(t) = t^2$, $y(t) = 4t$. Find its speed at $t = 3$.

$\frac{dx}{dt} = 2t$, $\frac{dy}{dt} = 4$. Speed $= \sqrt{(2t)^2 + 4^2}$; at $t = 3$: $\sqrt{36 + 16} = \sqrt{52} = 2\sqrt{13}$.

- 11 Find the arc length of $x = \cos^3 t$, $y = \sin^3 t$ for $0 \leq t \leq \frac{\pi}{2}$ (an astroid arc), using $\frac{dx}{dt} = -3\cos^2 t \sin t$ and $\frac{dy}{dt} = 3\sin^2 t \cos t$.

$\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 = 9\cos^4 t \sin^2 t + 9\sin^4 t \cos^2 t = 9\sin^2 t \cos^2 t (\cos^2 t + \sin^2 t) = 9\sin^2 t \cos^2 t$. So the speed is $3|\sin t \cos t|$. $L = \int_0^{\pi/2} 3\sin t \cos t dt = \left[\frac{3}{2}\sin^2 t\right]_0^{\pi/2} = \frac{3}{2}$.